

NCERT Exemplar Solutions for Class 12

Physics

Chapter 5 - Magnetism and Matter

Multiple Choice Question I

5.1 A toroid of n turns, mean radius R and cross-sectional radius a carries current I . It is placed on a horizontal table taken as x - y plane. Its magnetic moment m

- (a) is non-zero and points in the z -direction by symmetry.**
- (b) points along the axis of the toroid ($m = m\phi$).**
- (c) is zero, otherwise there would be a field falling as $\frac{1}{r^3}$ at large distances outside the toroid.**
- (d) is pointing radially outwards.**

Ans: Option (c) is correct. As there isn't a magnetic field outside of the toroid.

Outside, the M.F. decreases very quickly because it is inversely related to the third power of the distance from the toroid's centre.

Because there is no current, there is no magnetic moment outside, according to Amperian circuital law.

5.2 The magnetic field of Earth can be modelled by that of a point dipole placed at the centre of the Earth. The dipole axis makes an angle of 11.3° with the axis of Earth. At Mumbai, declination is nearly zero. Then,

- (a) the declination varies between 11.3° W to 11.3° E.**
- (b) the least declination is 0° .**

(c) the plane defined by dipole axis and Earth axis passes through Greenwich.

(d) declination averaged over Earth must be always negative.

Ans: Option (a) is correct because the magnetic field of the Earth is identical to that of a hypothetical magnetic dipole positioned at the centre of the Earth. The dipole's axis is inclined with respect to the earth's axis by on both the east and west sides (South pole of earth towards West and North pole towards east). As a result, the declination changes from East to West on both sides of throughout the whole earth's surface.

5.3 In a permanent magnet at room temperature

(a) magnetic moment of each molecule is zero.

(b) the individual molecules have non-zero magnetic moment which are all perfectly aligned.

(c) domains are partially aligned.

(d) domains are all perfectly aligned.

Ans: Option (c) is correct because, a permanent magnet behaves like a ferromagnetic material at ambient temperature. When there is no magnetic field, domains in ferromagnetic substances are randomly distributed.

As a result, the resulting magnetic moment is close to zero.

However, when this material is put in a strong magnetic field, certain domains remain aligned in the external field even when the external magnetic field is removed.

5.4 Consider the two idealized systems: (i) a parallel plate capacitor with large plates and small separation and (ii) a long solenoid of length $L \gg R$, radius of cross-section. In (i) E is ideally treated as a constant between plates and zero outside. In (ii) magnetic field is constant inside the solenoid and zero outside. These idealised assumptions, however, contradict fundamental laws as below:

(a) case (i) contradicts Gauss's law for electrostatic fields.

(b) case (ii) contradicts Gauss's law for magnetic fields.

(c) case (i) agrees with $\oint \mathbf{E} \cdot d\mathbf{l} = 0$.

(d) case (ii) contradicts $\oint \mathbf{H} \cdot d\mathbf{l} = I_{en}$

Ans: Option (b) is correct because,

According to Gauss's rule of electrostatic field $\oint_s \mathbf{E} \cdot d\mathbf{s} = \frac{q}{\epsilon_0}$

As the electric field lines do not form a continuous route, it does not contradict.

Gauss's magnetic field law states that $\oint_s \mathbf{B} \cdot d\mathbf{s} = 0$.

It contradicts for magnetic field since there is magnetic field inside the solenoid and no field outside solenoid carrying current, yet magnetic field lines create closed channels.

5.5 A paramagnetic sample shows a net magnetisation of 8Am^{-1} when placed in an external magnetic field of 0.6 T at a temperature of 4 K. When the same sample is placed in an external magnetic field of 0.2 T at a temperature of 16 K, the magnetisation will be

(a) $\frac{32}{3}\text{Am}^{-1}$

(b) $\frac{2}{3}\text{Am}^{-1}$

(c) 6Am^{-1}

(d) 2.4Am^{-1}

Ans: Option (b) is correct because Curie's equation of magnetization states that the magnetization of a substance is directly proportional to magnetic field induction (B) and inversely proportional to absolute temperature t,

$$I \propto \frac{B}{T} \text{ or } \frac{I_2}{I_1} = \frac{B_2 T_1}{B_1 T_2}$$

$$I_1 = 8 \text{ Am}^{-1}$$

$$B_1 = 0.6 \text{ T}$$

$$T_1 = 4 \text{ kI}_2 = ?$$

$$B_2 = 0.2 \text{ T}$$

$$T_2 = 16 \text{ k}$$

$$I_2 = \frac{B_2 T_1}{B_1 T_2}$$

$$I_1 = \frac{0.2 \times 8 \times 4}{0.6 \times 16} = \frac{2}{3} \text{ Am}^{-1}$$

Multiple Choice Question II

5.6 S is the surface of a lump of magnetic material.

- (a) Lines of B are necessarily continuous across S.**
- (b) Some lines of B must be discontinuous across S.**
- (c) Lines of H are necessarily continuous across S.**
- (d) Lines of H cannot all be continuous across S.**

Ans: Options (a) and (d) are correct, because magnetic field lines for magnetic induction (B) create continuous lines, lines of B must be continuous throughout S.

Also, the magnetic strength (H) to magnetise changes depending on whether the lump is within or outside.

As a result, the line of H cannot be continuous over S.

5.7 The primary origin(s) of magnetism lies in

- (a) atomic currents.**

(b) Pauli exclusion principle.

(c) polar nature of molecules.

(d) intrinsic spin of electron.

Ans: Options (a) and (d) are correct, Magnetism is produced by the motion of a charge particle, and the type of magnetism is determined by the nature of motion of the charge particle. In an atom, electrons orbit and spin around the nucleus, producing electricity and, as a result of the magnetic effect of the current, magnetism in the material.

5.8 A long solenoid has 1000 turns per metre and carries a current of 1A. It has a soft iron core of $\mu_r = 1000$. The core is heated beyond the Curie temperature, T_c .

(a) The H field in the solenoid is (nearly) unchanged but the B field decreases drastically.

(b) The H and B fields in the solenoid are nearly unchanged.

(c) The magnetisation in the core reverses direction.

(d) The magnetisation in the core diminishes by a factor of about 10^8 .

Ans: Options (a) and (d) are correct because,

$$n = 1000 \text{ turns/m}$$

$$\mu_r = 1000$$

$$H = nI = 1000 \times 1 = 1000 \text{ Amp}$$

It means H is static. Which shows option (a) is absolutely correct.

$$B = \mu_0 \mu_r nI = (\mu_0 nI) \mu_r = K \mu_r \text{ (K = constant)}$$

$$B \propto \mu_r.$$

When a ferromagnetic substance is heated above the Curie temperature, it acts like a paramagnetic substance, according to Curie's law.

Where,

$$(X_m)_{\text{ferro}} = 10^3$$

$$(X_m)_{\text{para}} = 10^{-5}$$

$$\therefore \frac{B_2}{B_1} = \frac{x_2}{x_1} = \frac{10^{-5}}{10^3} = 10^{-8} \text{ or } B_2 = 10^{-8} B_1$$

As a result, magnetization increases by 10^{-8} times or decreases by 10^8 times.

5.9 Essential difference between electrostatic shielding by a conducting shell and magnetostatic shielding is due to

- (a) electrostatic field lines can end on charges and conductors have free charges.**
- (b) lines of B can also end but conductors cannot end them.**
- (c) lines of B cannot end on any material and perfect shielding is not possible.**
- (d) shells of high permeability materials can be used to divert lines of B from the interior region.**

Ans: Options (a), (c) and (d) are correct, because conductors have free charge particles, they may halt lines of force, providing a shielding effect. Also monopole magnetic field lines do not exist, they cannot be halted or shielded.

Magnetic field lines are impacted by magnetic material and can be resisted by utilising a magnetic material with a high permeability to provide a zone with no magnetic field.

5.10 Let the magnetic field on earth be modelled by that of a point magnetic dipole at the centre of earth. The angle of dip at a point on the geographical equator

- (a) is always zero.**
- (b) can be zero at specific points.**
- (c) can be positive or negative.**

(d) is bounded.

Ans: Options (b), (c) and (d) are correct because the dipole is positioned at an angle of to from the geographical N-S axis.

Its south and north poles are analogous to a fictitious earth magnet.

The south pole is in the north direction towards west, while the north pole is in the south direction towards east.

The resultant magnetic field of a dipole will be zero in its equatorial plane but not on the geographical equator, proving the answer (b). When the equatorial plane of the dipole is changed, the angle of dip changes, i.e., where the angle of dip is zero. It does not equal 0 at all sites along the geographic equator. The dip angle is zero. It does not equal 0 at all sites along the geographic equator. Where the geographical equator meets the magnetic equator, the angle of dip is zero. As a result, solution (c) is verified, and (d).

VSA

5.11 A proton has spin and magnetic moment just like an electron. Why then its effect is neglected in magnetism of materials?

Ans: The magnetic moment of a charged particle with charge e and mass m ,

$$M_e = \frac{eh}{4\pi m_e} \text{ and } M_p = \frac{eh}{4\pi m_p}$$

Because the charges on the proton and electron are equivalent in size,

$$M \propto \frac{1}{m} \text{ or } \frac{M_e}{M_p} = \frac{m_p}{m_e}.$$

Because the mass of a proton is 1836 times that of an electron, the magnetic moment of a proton is

$$M_p = \frac{M_e}{m_p} m_e = \frac{M_e \cdot m_e}{1836 m_e}, M_p = \frac{1}{1836} \text{ of } M_e$$

As the magnetic moment of a proton is $1/1836$ times that of an electron, it may be ignored.

5.12 A permanent magnet in the shape of a thin cylinder of length 10 cm has $M = 10^6$ A / m . Calculate the magnetisation current I_M .

Ans: $M = 10^6$ ampere / metre , $l = 10\text{ cm} = 0.10\text{ m}$

I_M = Magnetisation current

$$M = \frac{I_M}{l}$$

Hence, $I_M = M \times l = 10^6 \times 0.1 = 10^5$ ampere.

5.13 Explain quantitatively the order of magnitude difference between the diamagnetic susceptibility of N_2 ($\sim 5 \times 10^{-9}$) (at STP) and Cu ($\sim 10^{-5}$).

Ans: By solving

$$\rho_N = \frac{28g}{22.4\text{ lit}} = \frac{28g}{22400\text{ ml}} = \frac{28}{22400} \text{ g per cm}^3$$

$$\rho_{cu} = 8 \text{ g / cm}^3$$

$$\therefore \frac{\rho_N}{\rho_{Cu}} = \frac{28}{22400 \times 8} = 1.6 \times 10^{-4}$$

$$\frac{\chi_N}{\chi_{Cu}} = \frac{5 \times 10^{-9}}{10^{-5}} = 5 \times 10^{-4}$$

$$\chi = \frac{\text{Intensity of magnetisation (M)}}{\text{Magnetising field intensity (H)}}$$

Magnetic moment (m)

$$x = \frac{\text{Volume (V)}}{H}$$

$$x = \frac{m}{HV}$$

Let, $m' = \text{mass}$

$V = \text{Velocity}$

$\rho = \text{density}$

Then,

$$\frac{m}{\rho} = V$$

$$x = \frac{m \times \rho}{Hm'}$$

As $x \propto \rho$

$$\frac{x_U}{x_{Cu}} = \frac{\rho_x}{\rho_{Cu}} = 1.6 \times 10^{-4}$$

The main difference in diamagnetic susceptibility of N_2 gas and solid Cu is related to density differences.

5.14 From molecular view point, discuss the temperature dependence of susceptibility for diamagnetism, paramagnetism and ferromagnetism.

Ans: Because the direction of the external magnetic field H and magnetism M caused by the orbital motion of electrons in a diamagnetic material are opposed, net magnetism is zero. As a result, temperature has little effect on the susceptibility x of diamagnetism.

Because the direction of magnetism caused by orbital motion of electrons in paramagnetism and ferromagnetism materials and the direction of the external

applied field are in the same direction, net magnetism increases and is greatly impacted by temperature.

The alignment of atomic magnetism is altered when temperature rises, resulting in a reduction in susceptibility.

5.15 A ball of superconducting material is dipped in liquid nitrogen and placed near a bar magnet. (i) In which direction will it move? (ii) What will be the direction of it's magnetic moment?

Ans: Liquid nitrogen and a superconducting material are diamagnetic materials, and the superconducting material remains diamagnetic after being immersed in liquid nitrogen. When an external magnetic field is applied to a superconducting material that has been dipped in liquid nitrogen, the external magnetic field repels the material, and the direction of motion is opposite to the direction of the magnet or magnetic field.

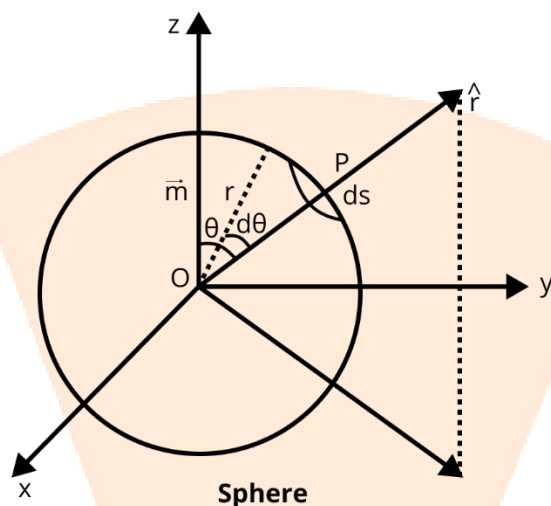
Short Answer Type Question

5.16 Verify the Gauss's law for magnetic field of a point dipole of dipole moment $\vec{m}$ at the origin for the surface which is a sphere of radius R .

Ans: Applying Gauss's Law,

$$\oint_s \vec{B} \cdot \vec{ds} = 0$$

The magnetic moment, $\vec{m}$ of a dipole at its origin O , $\vec{m} = m\hat{k}$



$$B = \frac{\mu_0}{4\pi} \frac{2\vec{m} \cos \theta}{r^3} \hat{r}$$

$$\vec{ds} = r(r \sin \theta) d\theta \hat{r} = r^2 \sin \theta \cdot d\theta \hat{r} \left[\because d\theta = \frac{ds}{r^2} \text{ or } ds = r^2 d\theta \right]$$

$$\begin{aligned} \oint_s \vec{B} \cdot \vec{ds} &= \oint \frac{\mu_0}{4\pi} \frac{2\vec{m} \cos \theta}{r^3} \hat{r} \cdot (r^2 \sin \theta \cdot d\theta \hat{r}) \\ &= \frac{\mu_0 \vec{m}}{4\pi r} \int_0^{2\pi} 2 \sin \theta \cos \theta \cdot d\theta = \frac{\mu_0 \vec{m}}{4\pi r} \int_0^{2\pi} \sin 2\theta \cdot d\theta \\ &= \frac{\mu_0 \vec{m}}{4\pi r} \left[\frac{-\cos 2\theta}{2} \right]_0^{2\pi} \\ &= \frac{\mu_0}{4\pi r \cdot 2} [\cos 4\pi - (-\cos 0)] \\ &= \frac{\mu_0}{4\pi r \times 2} [-\cos 0 + 1] \\ &= \frac{\mu_0}{8\pi r} [-1 + 1] \end{aligned}$$

$$\oint B \cdot ds = 0$$

It is proved.

5.17 Three identical bar magnets are rivetted together at centre in the same plane as shown in Fig. 5.1. This system is placed at rest in a slowly varying magnetic field. It is found that the system of magnets does not show any motion. The north-south poles of one magnet is shown in the Fig. 5.1. Determine the poles of the remaining two.

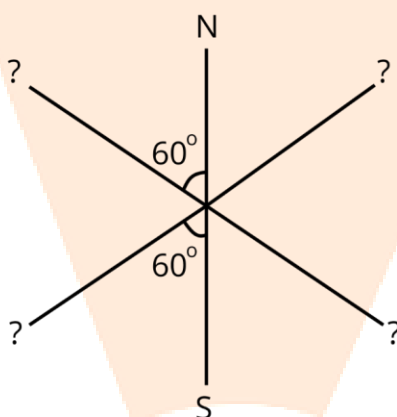
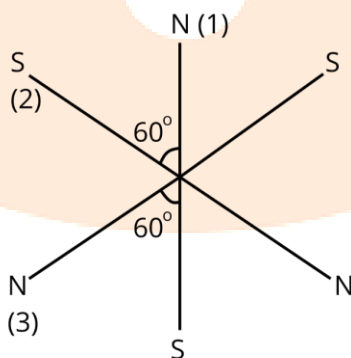


Fig. 5.1

Ans: The poles must be symmetric to each other otherwise a magnet will be formed.

It is only conceivable if the remaining two magnets' poles are arranged as shown in the illustration.



The north pole of magnet (1) is equally attracted by the south poles of magnets (2) and (3) arranged at equal distances apart.

Similarly, one pole of any one magnet is attracted by the opposing poles of the other two magnets, hence the resultant force or moment of each magnet is zero and it will not move when put on the table.

5.18 Suppose we want to verify the analogy between electrostatic and magnetostatic by an explicit experiment. Consider the motion of (i) electric dipole $\vec{p}$ in an electrostatic field $\vec{E}$ and (ii) magnetic dipole $\vec{m}$ in a magnetic field $\vec{B}$. Write down a set of conditions on $\vec{E}$, $\vec{B}$, $\vec{p}$, $\vec{m}$ so that the two motions are verified to be identical. (Assume identical initial conditions.)

Ans: Assume that angle between $\vec{m}$ and $\vec{B}$ is θ .

In a magnetic field, the torque on a magnetic dipole $\vec{B}$,

$$\tau = \vec{m} \vec{B} \sin \theta \dots\dots I$$

Just like above,

$$\tau' = \vec{p} \vec{E} \sin \theta \dots\dots II$$

Because if the velocity in I and II of an electric and magnetic dipole is similar,

$$\tau' = \tau$$

$$\vec{p} \vec{E} \sin \theta = \vec{m} \vec{B} \sin \theta$$

$$\vec{p} \vec{E} = \vec{m} \vec{B} \dots\dots III$$

$$\vec{E} = \vec{B} \vec{c} \text{ (relation between } \vec{E} \text{ and } \vec{B} \text{) } \dots\dots IV$$

In III, substitute the value of $\vec{E}$ from IV.

$$\vec{p} \vec{c} \vec{B} = \vec{m} \vec{B}$$

$$\vec{p} = \frac{\vec{m}}{c}$$

5.19 A bar magnet of magnetic moment m and moment of inertia I (about centre, perpendicular to length) is cut into two equal pieces, perpendicular to length. Let T be the period of oscillations of the original magnet about an axis through the mid point, perpendicular to length, in a magnetic field B . What would be the similar period T' for each piece?

Ans: If a magnetic moment magnet is split into n equal pieces,

The magnetic moment of all equal portions is equal to $nm' = m$.

As a result, the magnetic moment of each of the two components of the magnet

$$= m' = \frac{m}{2}.$$

$$I = \frac{ml^2}{12}$$

$$\text{Length of new magnet, } l' = \frac{l}{2}$$

$$\text{Actual time period, } T = 2\pi\sqrt{\frac{I}{mB}}$$

If M is the mass of the original magnet, then mass of each of the two magnets m' is $\frac{M}{2}$.

$$\frac{T}{T'} = \frac{2\pi\sqrt{\frac{I}{mB}}}{2\pi\sqrt{\frac{I'}{m'B}}} = \sqrt{\frac{I}{m} \cdot \frac{m'}{I'}} \quad \text{or} \quad \frac{T}{T'} = \sqrt{\frac{m}{m'} \cdot \frac{I'}{I}}$$

$$\frac{I'}{I} = \frac{\frac{m'l'^2}{12}}{\frac{ml^2}{12}} = \frac{m}{2} \cdot \left(\frac{1}{2}\right)^2$$

$$\frac{I'}{I} = \frac{m}{2} \cdot \frac{1}{4} = \frac{m}{8}$$

$$\frac{I'}{I} = \frac{\frac{m}{2} \cdot \frac{l^2}{4}}{ml^2} = \frac{1}{8}$$

$$\frac{m}{m'} = \frac{m}{\frac{m}{2}} = \frac{2}{1}$$

$$\therefore \frac{T}{T'} = \sqrt{\frac{2}{1} \times \frac{1}{8}} = \sqrt{\frac{1}{4}} \frac{T}{T'} = \frac{1}{2} \text{ or}$$

$$T' = \frac{T}{2} \text{ sec}$$

5.20 Use (i) the Ampere's law for $\vec{H}$ and (ii) continuity of lines of $\vec{B}$, to conclude that inside a bar magnet, (a) lines of $\vec{H}$ run from the N pole to S pole, while (b) lines of $\vec{B}$ must run from the S pole to N pole.

$$\text{Ans: } \int_P^Q \vec{H} \cdot d\vec{l} = \int_Q^P \frac{\vec{B}}{\mu_0} d\vec{l}$$

Where B is magnetic field and m_0 is dipole moment. As angle between B and dl varies from 90° , $0, 90^\circ$ from R to T , so $\cos \theta$ is greater than 1.

$$\text{So } \int_P^Q \vec{H} \cdot d\vec{l} = \int_Q^P \frac{\vec{B}}{\mu_0} d\vec{l} > 0 \text{ i.e.}$$

Positive.

As a result, the value of B must be adjusted inside the magnet from south pole to north pole.

Applying Ampere's Law,

$$\oint_{PQP} \vec{H} \cdot d\vec{l} = 0$$

$$\oint_{PQP} \vec{H} \cdot d\vec{l} = \int_P^Q \vec{H} \cdot d\vec{l} + \int_Q^P \vec{H} \cdot d\vec{l} = 0$$

Because $\int_P^0 H \cdot dl > 0$ is outer the magnet,

$\int_Q^P H \cdot dl < 0$ is inner the magnet, Because the angle between H and dl is greater than 90° within the magnet, $\cos\theta$ is negative.

This means that the H lines must travel from north pole to south pole.

Long Answer Type Questions

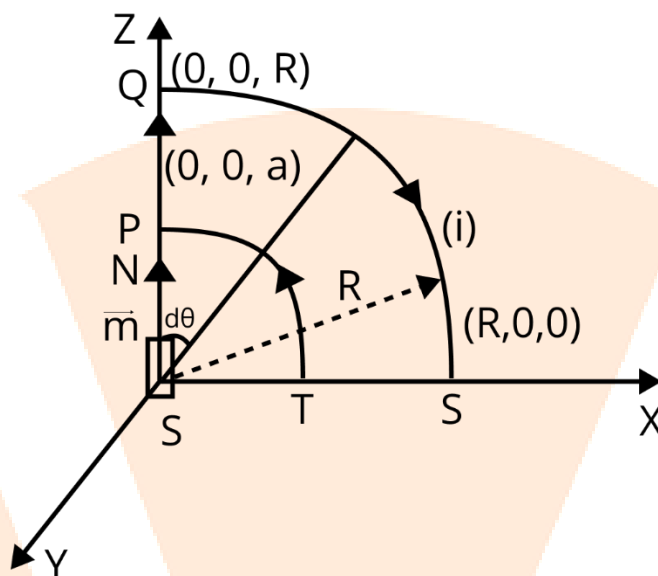
5.21 Verify the Ampere's law for magnetic field of a point dipole of dipole moment $\vec{m} = m\hat{k}$. Take C as the closed curve running clockwise along

(i) the z-axis from $z = a > 0$ to $z = R$;

Ans: In the picture from P to Q, every point on the z axis that is on a dipole is in the $\hat{k}$ direction.

Hence every point on the Z-axis that is on an axial dipole NS is positioned at the origin.

$$\text{So, } |B| = \frac{\mu_0}{4\pi} \frac{2(\vec{m})}{z^3} = \frac{\mu_0 \vec{m}}{2\pi z^3}$$

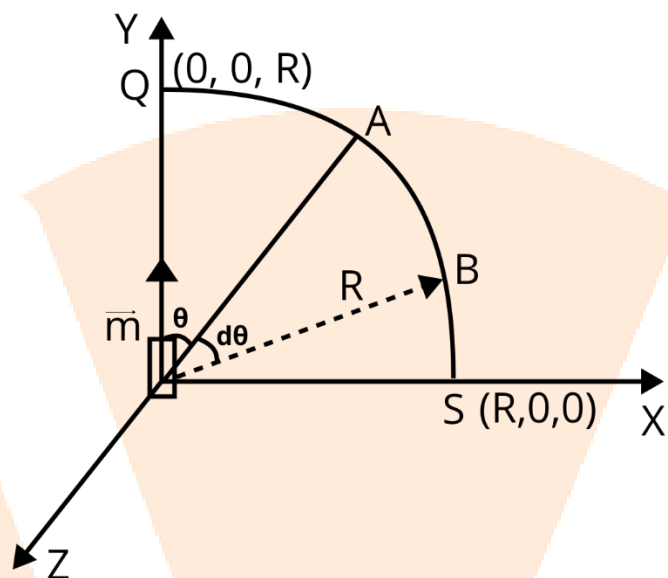


$$\begin{aligned}
 \oint_P \vec{B} \cdot d\vec{l} &= \int_P^Q B dl \cos 0^\circ = \int_P^Q B dz \\
 &= \int_P^Q \frac{\mu_0 \vec{m}}{2\pi z^3} dz = \frac{\mu_0 m}{2\pi} \int_a^R z^{-3} dz \\
 &= \frac{\mu_0 \vec{m}}{2\pi} \left[\frac{z^{-2}}{-2} \right]_a^R = \frac{\mu_0 \vec{m}}{2\pi(-2)} \left[\frac{1}{R^2} - \frac{1}{a^2} \right] \\
 \int_P B \cdot dl &= \frac{\mu_0 \vec{m}}{4\pi} \left[\frac{1}{a^2} - \frac{1}{R^2} \right]
 \end{aligned}$$

(ii) along the quarter circle of radius R and centre at the origin, in the first quadrant of x-z plane;

Ans: Ampere's law is illustrated in the image here along the quarter circle QS of radius R.

Point A can be thought of as being on the equatorial line of a magnetic dipole of moment $\vec{m} \sin \theta$.



$$B = \frac{\mu_0}{4\pi} \frac{\vec{m} \sin \theta}{R^3}$$

$$d\theta = \frac{dl}{R} \Rightarrow dl = R \cdot d\theta$$

Applying Ampere's Law,

$$\int \vec{B} \cdot d\vec{l} = \int B \cdot dl \cos \theta$$

$$\int \vec{B} \cdot d\vec{l} = \int_0^{\pi/2} \frac{\mu_0}{4\pi} \frac{\vec{m} \sin \theta}{R^3} \cdot R d\theta$$

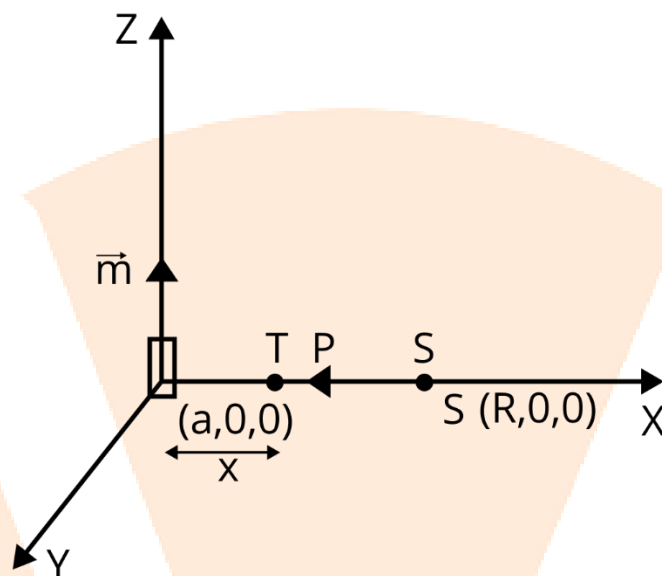
$$\int \vec{B} \cdot d\vec{l} = \frac{\mu_0 \vec{m}}{4\pi R^2} [-\cos \theta]_0^{\pi/2}$$

$$= \frac{\mu_0 \vec{m}}{4\pi R^2} [-\cos 90^\circ + \cos 0^\circ]$$

$$\int \vec{B} \cdot d\vec{l} = \frac{\mu_0 \vec{m}}{4\pi R^2}$$

(iii) along the x-axis from $x = R$ to $x = a$, and

Ans:



Because all locations from S to T lie on the equatorial line of the magnetic dipole N-S, magnetic field induction occurs at a point P at a distance x from the dipole.

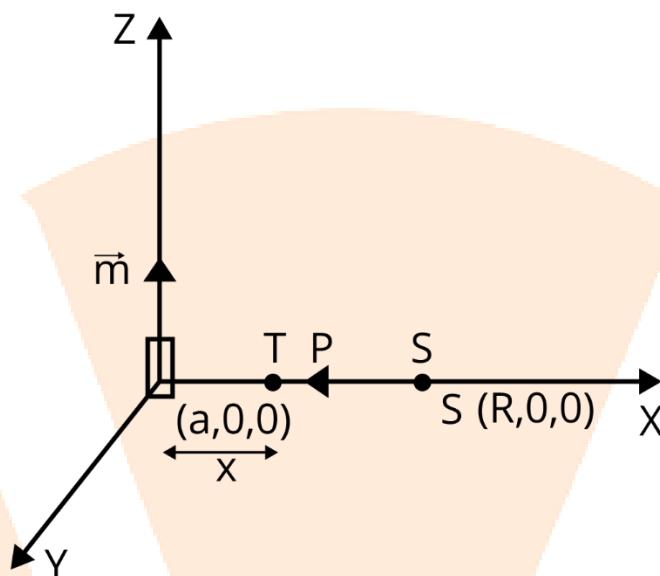
$$B = \frac{\mu_0}{4\pi} \frac{\vec{m}}{x^3} = \frac{\mu_0}{4\pi} \frac{\vec{m}\hat{k}}{x^3}$$

$$\int \vec{B} \cdot \vec{dl} = \int_R^a \frac{-\mu_0 \vec{m}\hat{k}}{4\pi x^3} \cdot d\vec{l} \because \text{angle between } d\vec{l} \text{ and } \vec{m} \text{ is } 90^\circ$$

$$\int \vec{B} \cdot \vec{dl} = \int_R^a \frac{-\mu_0 |\vec{m}| dl \cos 90^\circ}{4\pi x^3} = 0$$

(iv) along the quarter circle of radius a and centre at the origin in the first quadrant of x - z plane.

Ans:



$$\int B \cdot dl = \int_{\theta=2}^0 \frac{\mu_0}{4\pi} \frac{\vec{m} \sin \theta}{a^3} \cdot a d\theta$$

$$\int B \cdot dl = \frac{\mu_0 \vec{m}}{4\pi a^2} [-\cos \theta]_{\pi/2}^0$$

θ = angle from axial line of dipole n

$$\int B \cdot dl = \frac{\mu_0 \vec{m}}{4\pi a^2} \left[-\cos 0 + \cos \frac{\pi}{2} \right] = \frac{\mu_0 \vec{m}}{4\pi a^2} (-1 + 0) = \frac{-\mu_0 \vec{m}}{4\pi a^2}$$

Applying Ampere's Law,

$$\oint_{PQST} B \cdot dl = \int_P^Q \vec{B} \cdot d\vec{l} + \int_Q^S \vec{B} \cdot d\vec{l} + \int_S^T \vec{B} \cdot d\vec{l} + \int_T^P B \cdot dl$$

Using (i), (ii), (iii) values,

$$\oint_{PQST} B \cdot dl = \frac{\mu_0 \vec{m}}{4\pi} \left(\frac{1}{a^2} - \frac{1}{R^2} \right) + \frac{\mu_0 \vec{m}}{4\pi R^2} + 0 + \frac{-\mu_0 \vec{m}}{4\pi a^2}$$

$$= \frac{\mu_0 \vec{m}}{4\pi a^2} - \frac{\mu_0 \vec{m}}{4\pi R^2} + \frac{\mu_0 \vec{m}}{4\pi R^2} - \frac{\mu_0 \vec{m}}{4\pi a^2}$$

$$\oint_{PQST} B \cdot dl = 0$$

Now, it is proved.

5.22 What are the dimensions of χ , the magnetic susceptibility? Consider an H-atom. Guess an expression for χ , upto a constant by constructing a quantity of dimensions of χ , out of parameters of the atom: e , m , v , R and μ_0 . Here, m is the electronic mass, v is electronic velocity, R is Bohr radius. Estimate the number so obtained and compare with the value of $|\chi| \sim 10^{-5}$ for many solid materials.

Ans: Because magnetization intensity (M) and magnetising field (H) have the same unit, i.e. ampere/metre and $\chi = \frac{M}{H}$. As a result, susceptibility has no unit.

Using Biot-Savart's Law,

$$dB = \frac{\mu_0}{4\pi} = \frac{I \cdot dl \sin \theta}{r^2}$$

$$F = Bqv \sin \theta$$

$$= \frac{F}{qv \sin \theta}$$

$$= \frac{MLT^{-2}}{QLT^{-1}}$$

$$= [ML^0T^{-1}Q^{-1}]$$

$$\therefore \mu_0 = \frac{[MT^1Q^{-1}]L^2}{QT^{-1}L}$$

$$= [MLQ^{-2}]$$

χ is affected by the magnetic moment created when H is turned on.

Through its charge e , H bonds to atomic electrons.

The influence on m is caused by current I , which involves another e -factor.

$$\chi = \mu_0^a e^2 m^b v^c R^d$$

$$[M^0 L^0 T^0 Q^0] = [MLQ^{-2}]^a Q^2 M^b [LT^{-1}]^c [L]^d$$

$$C = 0, a + b = 0 - 2a + 2 = 0 \quad a + c + d = 0$$

$$1 + b = 0 - 2a = -21 + 0 + d = 0$$

$$b = -1a = +1d = -1$$

$$\chi = (\mu_0) e^2 m^{-1} v^0 R^{-1}$$

$$\chi = \frac{\mu_0 e^2}{mR}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ Tm A}^{-1} \text{ e} = 1.6 \times 10^{-19}$$

$$m = 9.1 \times 10^{-31} \text{ kg} \quad R = 10^{-10} \text{ m}$$

$$\chi = \frac{(4\pi \times 10^{-7})(1.6 \times 10^{-19})(1.6 \times 10^{-19})}{(9.1 \times 10^{-31}) \times 10^{-10}}$$

$$= \frac{4 \times 3.1 \times 1.6 \times 1.6}{9.1} \times 10^{-7-19-19+31+10}$$

$$= \frac{124 \times 256 \times 10^{-45+41}}{9.1} = \frac{317.4}{91} \times 10^{-4}$$

$$= 3.5 \times 10^{-4} |\chi'| = 10^{-5} \text{ (given)}$$

$$\frac{\chi}{|\chi'|} = \frac{3.5 \times 10^{-4}}{10^{-5}} = \frac{3.5 \times 10^{-4}}{10^{-1} \times 10^{-4}} = 35$$

$$\chi = 35 |\chi'|$$

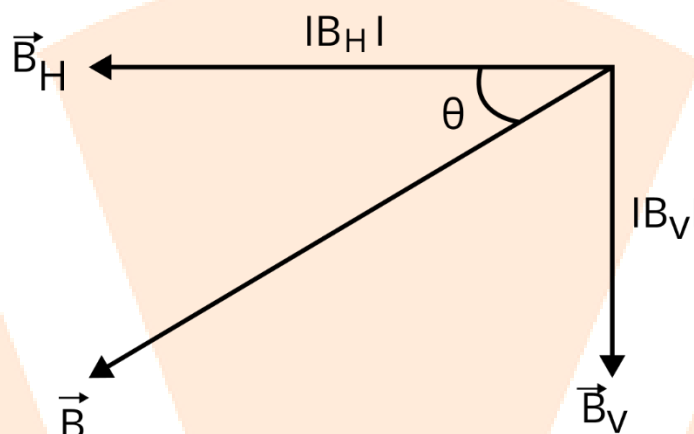
5.23 Assume the dipole model for earth's magnetic field B which is given by

$$B_v = \text{vertical component of magnetic field} = \frac{\mu_0}{4\pi} \frac{2m \cos \theta}{r^3} \quad B_H = \text{Horizontal}$$

component of magnetic field = $\frac{\mu_0}{4\pi} \frac{\sin \theta m}{r^3}$ $\theta = 90^\circ$ - l attitude as measured from magnetic equator. Find loci of points for which

(i) |B| is minimum;

Ans:



$$B^2 + B_V^2 + B_H^2$$

$$= \left[\frac{\mu_0}{4\pi} \frac{2m \cos \theta}{r^3} \right]^2 + \left[\frac{\mu_0}{4\pi} \frac{m \sin \theta}{r^3} \right]^2$$

According to the question, using values of B_V and B_H .

$$= \left[\frac{\mu_0}{4\pi} \right]^2 \frac{m^2}{r^6} [4 \cos^2 \theta + \sin^2 \theta] B^2$$

$$= \left(\frac{\mu_0}{4\pi} \right) \times \frac{m^2}{r^6} [3 \cos^2 \theta + 1] B$$

$$= \frac{\mu_0}{4\pi} \frac{m}{r^3} [3 \cos^2 \theta + 1]^{1/2} \dots (i)$$

Using equation (i), the value of B will be minimum when $[3 \cos^2 \theta + 1]^{1/2}$ is minimum which will be at $\theta = \frac{\pi}{2}$. So magnetic equator lies at $\theta = \frac{\pi}{2}$, i.e., from magnetic dipole axis $\theta = \frac{\pi}{2}$ for magnetic equator.

(ii) dip angle is zero; and

Ans: dip angle is

$$\tan \delta = \frac{B_V}{B_H} = \frac{\frac{\mu_0}{4\pi} \frac{2m \cos \theta}{r^3}}{\frac{\mu_0}{4\pi} \frac{m \sin \theta}{r^3}}$$

$$\tan \delta = 2 \cot \theta \text{ for}$$

$$\delta = 0, \cot \theta = 0, \theta = \frac{\pi}{2}$$

As a result, the angle of dip will be near the magnetic equator.

(iii) dip angle is $\pm 45^\circ$.

Ans: $\tan \delta = \frac{B_V}{B_H} = \text{angle of dip } \delta = \pm 45^\circ$

$$\frac{B_V}{B_H} = \tan \pm 45^\circ$$

$$\Rightarrow \frac{B_V}{B_H} = \tan 45^\circ$$

$$\frac{B_V}{B_H} = 1$$

$$B_V = B_H$$

$$\delta = \pm 45^\circ$$

$$\tan \pm 45^\circ = 2 \cot \theta [\because \tan \delta = 2 \cot \theta]$$

$$\cot \theta = \frac{1}{2}$$

$$\tan \theta = 2$$

$$\cot \theta = \frac{1}{2}$$

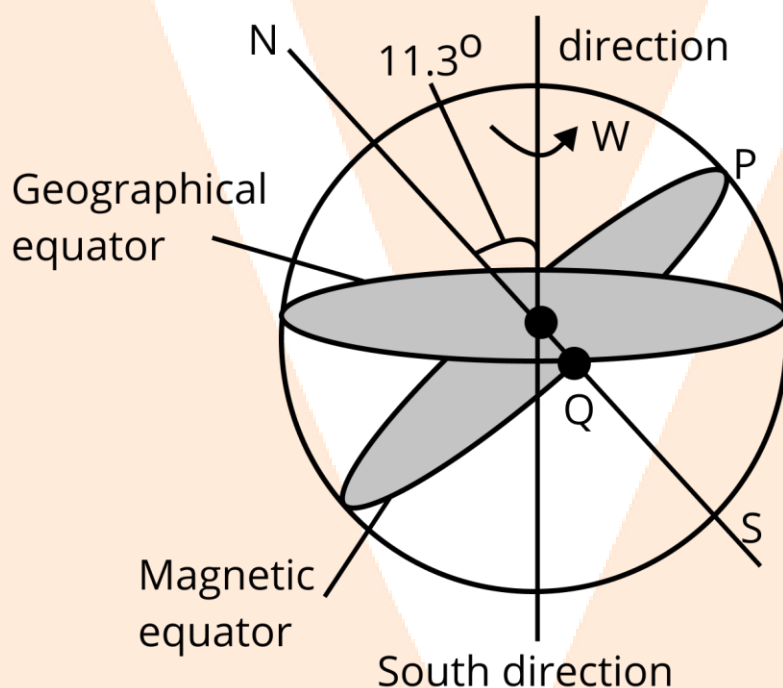
or $\tan \theta = 2 \Rightarrow \theta = \tan^{-1} 2$, here angle of dip is $\delta = \pm 45^\circ$

5.24 Consider the plane S formed by the dipole axis and the axis of earth. Let P be point on the magnetic equator and in S. Let Q be the point of intersection of the geographical and magnetic equators. Obtain the declination and dip angles at P and Q.

Ans: Q is the place at where the geographical and magnetic meridian meet.

As a result, the angle dip at P and Q will be 0 with horizontal as the magnetic mid point remains horizontal.

The angle formed between the rotation axis of the Earth and the magnetic axis is 11.3° and their equators are at 90° with their axes.



So the angle between the magnetic and geographical planes, i.e. declination, will be 11.3° at both P and Q.

5.25 There are two current carrying planar coils made each from identical wires of length L. C_1 is circular (radius R) and C_2 is square (side a). They are so constructed that they have same frequency of oscillation when they are placed in the same uniform B and carry the same current. Find a in terms of R.

Ans: Because the frequency ω for both coils are the same,

$$\omega_1 = \omega_2$$

$$\frac{2\pi}{T_1} = \frac{2\pi}{T}$$

So the time periods of the coils C_1 and C_2 are equal, so

$$T_1 = T_2$$

$$2\pi \sqrt{\frac{I_1}{m_1 B}} = 2\pi \sqrt{\frac{I_2}{m_2 B}}$$

$$\sqrt{\frac{I_1}{m_1}} = \sqrt{\frac{I_2}{m_2}}$$

Squaring both sides,

$$\frac{I_1}{I_2} = \frac{m_1}{m_2} \dots (i)$$

I_1, I_2 are the moment of inertia of coils C_1 and C_2 ,

$$I_1 = \frac{mR^2}{2} \dots (ii)$$

$$I_2 = \frac{ma^2}{12} \dots (iii)$$

Because wire has similar length, then mass will also be similar.

$$m_1 = n_1 I A_1$$

$$m_1 = \frac{L}{2\pi R} \cdot I \cdot \pi R^2 [\because L = 2\pi R n_1]$$

$$m_1 = \frac{LIR}{2} \dots (iv)$$

Magnetic moment for a square-shaped coil

$$m_2 = n_2 I A_2$$

$$m_2 = \frac{LIa}{4}$$

Putting all values in (i),

$$\Rightarrow \frac{\frac{mR^2}{12}}{\frac{2}{ma^2}} = \frac{\frac{LIR}{4}}{\frac{LIa}{4}}$$

$$\Rightarrow \frac{R^2}{2} \times \frac{12}{a^2} = \frac{R}{2} \times \frac{4}{a}$$

$$\Rightarrow \frac{R}{a} = \frac{1}{3}$$

$$\Rightarrow 3R = a$$